

Cardiovascular Disease — EDA Insights Report

Week 2 Internship Project · Data Science Track

1. Dataset Summary

This analysis is based on the Cardiovascular Disease dataset (Kaggle, sulianova), containing patient health records with demographic, physiological, and lifestyle attributes. The bundled sample file contains 1,512 records; the full published dataset contains 70,000 records. After cleaning (duplicate removal, missing-value imputation, and outlier treatment), 1,428 records remained for analysis.

2. Data Cleaning Summary

Step	Action	Rows affected
1	Removed exact duplicate rows	12
2	Converted age from days to years	All rows
3	Filled missing numeric values (median imputation)	24 values
4	Removed physiologically impossible blood pressure readings	15
5	Clipped extreme height/weight outliers (1st-99th pct)	57
6	Engineered BMI feature	All rows

3. Key Insights

- Age and systolic blood pressure (ap_hi) show the strongest correlation with cardiovascular disease among all variables analyzed, consistent with established clinical risk factors.
- Patients with well-above-normal cholesterol show a substantially higher disease rate than patients with normal cholesterol — roughly double in the filtered sample.
- Physically active patients show a meaningfully lower disease rate than inactive patients, supporting exercise as a protective factor.
- Disease rate increases gradually and consistently with age, with no single sharp threshold age at which risk jumps — risk appears cumulative rather than triggered at one point.
- BMI alone is a weaker predictor than blood pressure or cholesterol in this dataset, suggesting that direct physiological markers carry more predictive signal than body composition alone.

4. Recommendations

- Prioritize blood pressure and cholesterol screening as early, low-cost risk indicators, particularly for patients above age 45.
- Promote physical activity programs as a low-cost intervention, given the consistent association between activity and lower disease rates.
- Consider combined-risk screening (elevated cholesterol + low activity + age 50+) as a priority group for more intensive monitoring, since this profile shows compounded risk.

5. Technologies Used

Python, Pandas, NumPy, Plotly, Matplotlib, Seaborn, Streamlit. Full source code and interactive dashboard available in the project GitHub repository.